Empirical Investigation of Multi-Sensor Fusion, Representation Collapse, and Policy Stability in CARLA Reinforcement Learning

Bachelor Thesis Project (BTP)
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Project Codebase: https://github.com/MSR327/vlm

***Abstract*—Deep reinforcement learning (RL) for vision-based autonomous driving requires balancing perceptual coverage, state dimensionality, and policy optimization stability. We present a systematic empirical study on sensor scaling and representation learning in the CARLA simulator using Proximal Policy Optimization (PPO). Rather than reporting only an idealized endpoint, we evaluate an empirical ladder across four sensory configurations: (1) a monocular baseline (100-dim) achieving 39.0% route completion (195 m) in Town01 but failing at sharp intersections (Meter 190) and collapsing under zero-shot transfer in Town02 (1.0% completion, 100% collisions); (2) a naive 4-camera surround rig (385-dim) that induces policy collapse (3.4% completion) due to counter-directional optic flow and triggers DirectX 11 buffer overruns; (3) a pruned 3-camera forward rig (290-dim, $0^\circ, \pm 60^\circ$) recovering completion to 25.0% ($R = 158.31$); and (4) an orthogonal multimodal fusion rig (195-dim) pairing front vision with a 2D Bird's-Eye-View (BEV) LiDAR height grid, achieving monotonic reward progression ($-8.81 \rightarrow -5.02$, peak $+19.84$) with zero stalls. Furthermore, we isolate and resolve two severe numerical instabilities in sparse BEV autoencoding: a float32 exponential overflow (NaN loss) cured via bounded log-sigma clipping $[-15.0, 1.0]$, and a 1.8×10^6 latent variance collapse cured via unconditional forward execution. Finally, we analyze the structural limits of static latent concatenation across independent VAEs, proving that ungrounded MLP policy heads lack spatial cross-attention mechanisms necessary to reconcile multi-modal features, establishing a principled empirical motivation for patch-based Vision-Language-Action (VLA) and cross-attention architectures.**

***Index Terms*—Autonomous driving, CARLA simulator, sensor fusion, Bird's-Eye-View LiDAR, Proximal Policy Optimization, Variational Autoencoder, representation learning, cross-attention.**

I. Related Work

A. Sensor Fusion in Autonomous Driving

End-to-end autonomous driving maps multi-modal perceptual streams directly to low-level vehicular continuous control commands [1]–[3]. While foundational imitation learning frameworks operated primarily on monocular perspective RGB cameras, navigating dense urban environments demands broader spatial coverage and robust depth perception [4], [6]. *Learning by Cheating* (LBC) [6] proved that monocular camera agents suffer from severe spatial myopia, whereas agents trained on privileged Bird's-Eye-View (BEV) abstractions achieve substantially higher route completion. *TransFuser* [5] demonstrated that combining multi-view cameras with LiDAR point clouds via cross-attention outperforms single-modality baselines by fusing optical semantic segmentation with invariant metric 3D geometry. *InterFuser* [7] extended multi-sensor fusion across panoramic perspectives to eliminate blind spots and ensure collision safety in complex traffic scenarios.

B. Bird's-Eye-View (BEV) Point Cloud Representations

Projecting raw 3D LiDAR point clouds into 2D Bird's-Eye-View (BEV) spatial grids preserves metric physical distances while enabling standard 2D convolutional backbones to extract geometric affordances efficiently: *PointPillars* [8] pioneered vertical pillar tokenization for real-time 3D object detection from point clouds. *BEVDet* [9] further established that transforming perceptual features into an ego-centric 2D metric ground plane provides superior spatial localization and trajectory prediction compared to perspective-view processing.

C. Representation Learning & Dimensionality in RL

Model-free reinforcement learning directly from raw high-dimensional pixels suffers from extreme sample inefficiency, sample complexity, and optimization instability [10], [12]. *CARL* [10] demonstrated in CARLA benchmark evaluations that policy gradient convergence is heavily constrained by observation representation dimensionality. *Roach* [11] addressed high-dimensional sample inefficiency by pre-training policies over privileged BEV affordance masks. Variational Autoencoders (VAEs) [15] are widely deployed to compress high-dimensional observation frames into compact, regularized continuous latent vectors $\mathbf{z} \in \mathbb{R}^{d_z}$ to feed downstream policy heads. Furthermore, abstracting natural RGB streams into Cityscapes semantic class palettes [16] removes extraneous variance originating from dynamic weather conditions, sun glare, and high-frequency textures.

D. Policy Optimization & Control Stability

Proximal Policy Optimization (PPO) [12] represents the primary on-policy clipped surrogate reinforcement learning objective for continuous control. Generalized Advantage Estimation (GAE-λ) [13] introduces exponential discounting to calibrate the bias-variance trade-off in policy advantage estimates. Conditioning for Action Policy Smoothness (CAPS) [14] demonstrated that model-free continuous RL agents frequently suffer from high-frequency actuator chatter and aggressive steering jitter unless explicitly regularized via temporal action smoothing and smoothness penalties.

II. Methodology

A. Reinforcement Learning Formulation

We formulate the closed-loop autonomous driving problem as an episodic Partially Observable Markov Decision Process (POMDP) defined by the tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$:

- **Control Frequency:** The agent interacts synchronously with the CARLA simulator at a fixed frequency of $f = 20$ Hz ($\Delta t = 0.05$ s).
- **Action Space:** The policy outputs a continuous 2-dimensional action vector $\mathbf{a}_t = [a_{\text{steer}}, a_{\text{long}}] \in [-1, 1]^2$.
- **Decoupled Longitudinal Control:** To prevent simultaneous, conflicting throttle and brake commands, we implement a signed longitudinal mapping:

$$\text{throttle}_t = \max(a_{\text{long}},\ 0.0) \tag{1}$$
$$\text{brake}_t = \max(-a_{\text{long}},\ 0.0) \tag{2}$$

- **Temporal Action Smoothing:** Guided by the CAPS framework [14], we filter steering predictions using an Exponential Moving Average (EMA) to prevent vehicle instability and actuator chatter:

$$\delta_t = 0.6\,\delta_{t-1} + 0.4\,a_{\text{steer},t} \tag{3}$$

where δ_t denotes the physical steering angle applied to the CARLA ego-vehicle.

- **Multi-Objective Reward Function:** At each discrete time step t, the agent receives a shaped reward:

$$r_t = R_{\text{speed}} \cdot R_{\text{center}} \cdot R_{\text{angle}} + R_{\text{penalty}} \tag{4}$$

where the individual components are defined as:

$$R_{\text{speed}} = \min(v_t/22.0,\ 1.0) \tag{5}$$
$$R_{\text{center}} = \max\left(1.0 - \frac{d_{\text{center}}}{3.0},\ 0.0\right) \tag{6}$$
$$R_{\text{angle}} = \max\left(1.0 - \frac{|\theta_{\text{heading}}|}{20^\circ},\ 0.0\right) \tag{7}$$

Here v_t represents scalar forward speed in km/h, d_{center} denotes signed perpendicular lane-center deviation in meters, and θ_{heading} represents the angular heading error relative to the lane tangent. Dynamic collisions, sidewalk encroachments, or lane deviations exceeding 3.0 m terminate the episode with a penalty of $R_{\text{penalty}} = -10.0$.

B. The Empirical Sensor Ladder

To isolate the empirical trade-offs between perceptual coverage, representation dimensionality, and policy stability, we evaluate four systematic sensor rungs (Fig. 1):

1) **Rung 1: Monocular Baseline (100-dim):** A single front-facing semantic camera (0° yaw, 125° FOV). The state vector concatenates the 95-dimensional visual latent with a 5-dimensional navigation telemetry vector:

$$\mathbf{s}_t = [\mathbf{z}_{\text{cam}} \in \mathbb{R}^{95} \,\|\, \mathbf{x}_{\text{nav}} \in \mathbb{R}^{5}] \in \mathbb{R}^{100} \tag{8}$$

where $\mathbf{x}_{\text{nav}} = [v_t, d_{\text{center}}, \theta_{\text{heading}}, \delta_{t-1}, a_{\text{long},t-1}]$.

2) **Rung 2a: Naive 4-Camera Surround Rig (385-dim):** Four orthogonal semantic cameras covering the full horizontal perimeter: Front ($0^\circ, 125^\circ$ FOV), Left ($-90^\circ, 90^\circ$ FOV), Right ($+90^\circ, 90^\circ$ FOV), and Rear ($180^\circ, 90^\circ$ FOV). The state is formed by direct latent concatenation:

$$\mathbf{s}_t = [\mathbf{z}_{\text{front}} \,\|\, \mathbf{z}_{\text{left}} \,\|\, \mathbf{z}_{\text{right}} \,\|\, \mathbf{z}_{\text{rear}} \,\|\, \mathbf{x}_{\text{nav}}] \in \mathbb{R}^{385} \tag{9}$$

3) **Rung 2b: Pruned 3-Camera Forward Surround (290-dim):** To eliminate counter-directional optic flow from the rear camera while maintaining lateral intersection coverage, the rig is reconfigured to three forward-focused views: Front ($0^\circ, 125^\circ$ FOV), Left-Angled ($-60^\circ, 90^\circ$ FOV), and Right-Angled ($+60^\circ, 90^\circ$ FOV), yielding a combined panoramic forward arc of 210°:

$$\mathbf{s}_t = [\mathbf{z}_{\text{front}} \,\|\, \mathbf{z}_{\text{left}} \,\|\, \mathbf{z}_{\text{right}} \,\|\, \mathbf{x}_{\text{nav}}] \in \mathbb{R}^{290} \tag{10}$$

4) **Rung 3: Orthogonal Multimodal Fusion (195-dim):** One front-facing camera ($0^\circ, 125^\circ$ FOV) combined with a roof-mounted 3D LiDAR projected into a 2D Bird's-Eye-View (BEV) height/occupancy grid:

$$\mathbf{s}_t = [\mathbf{z}_{\text{cam}} \in \mathbb{R}^{95} \,\|\, \mathbf{z}_{\text{lidar}} \in \mathbb{R}^{95} \,\|\, \mathbf{x}_{\text{nav}} \in \mathbb{R}^{5}] \in \mathbb{R}^{195} \tag{11}$$

C. 2D Bird's-Eye-View (BEV) LiDAR Projection Geometry

A 360-degree raycast LiDAR sensor is top-mounted on the ego-vehicle chassis at ($x = 0.0, y = 0.0, z = 2.4$ m). At each time step, raw 3D point cloud returns $\mathcal{P} = \{(x_i, y_i, z_i)\}_{i=1}^{N}$ are projected into an asymmetric Bird's-Eye-View spatial grid (Fig. 2):

1) **Asymmetric Spatial Bounding:** We define an asymmetric region of interest biased longitudinally forward:

$$\mathcal{P}_{\text{filt}} = \big\{(x_i, y_i, z_i) \in \mathcal{P} \mid -15.0 \le x_i \le +35.0,\\ -25.0 \le y_i \le +25.0\big\} \tag{12}$$

2) **Discretization to Pixel Grid Coordinates (u_i, v_i):** The continuous spatial coordinates are mapped into an 80×160 discrete matrix:

$$u_i = \text{clip}\left(\left\lfloor \frac{35.0 - x_i}{50.0} \times 79 \right\rfloor,\ 0,\ 79\right) \tag{13}$$
$$v_i = \text{clip}\left(\left\lfloor \frac{y_i + 25.0}{50.0} \times 159 \right\rfloor,\ 0,\ 159\right) \tag{14}$$

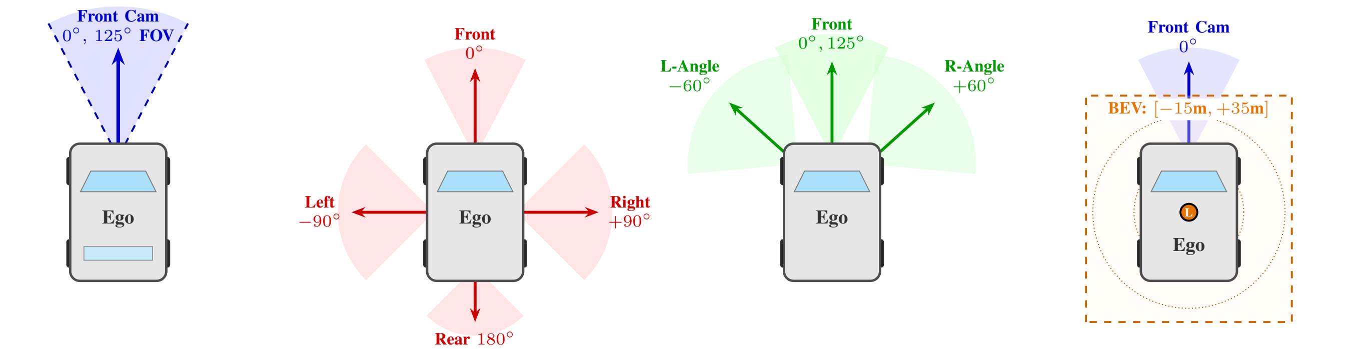
3) **Normalized Height Assignment:** The elevation $z_i \in [-2.0 \text{ m}, +2.0 \text{ m}]$ is mapped to continuous unit range:

$$\tilde{z}_i = \text{clip}\left(\frac{z_i + 2.0}{4.0},\ 0.0,\ 1.0\right) \tag{15}$$

4) **Max-Pooling Channel Assignment:** Where multiple LiDAR reflections project into the same spatial pixel (u, v), max-pooling preserves the highest obstacle return:

$$\mathbf{L}_{\text{bev}}(u, v) = \max_{i: u_i = u, v_i = v} \tilde{z}_i \tag{16}$$

yielding a single-channel raster $\mathbf{L}_{\text{bev}} \in [0, 1]^{80 \times 160}$.



(a) Rung 1: Monocular $\mathbf{s}_t \in \mathbb{R}^{100}$ (95c + 5n)

(b) Rung 2a: 4-Cam $\mathbf{s}_t \in \mathbb{R}^{385}$ (380c + 5n)

(c) Rung 2b: Pruned 3-Cam $\mathbf{s}_t \in \mathbb{R}^{290}$ (285c + 5n)

(d) Rung 3: Cam + LiDAR $\mathbf{s}_t \in \mathbb{R}^{195}$ (95c + 95L + 5n)

Fig. 1. Architectural diagrams of the four sensor ladder rigs evaluated in CARLA. (a) Rung 1: Monocular front baseline. (b) Rung 2a: Naive 4-camera surround rig. (c) Rung 2b: Pruned 3-camera forward rig (210° panoramic forward arc). (d) Rung 3: Orthogonal multimodal fusion of front camera and 2D BEV LiDAR.

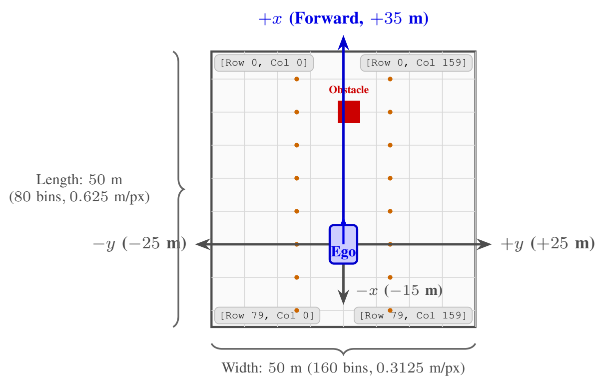


Fig. 2. Geometric coordinate mapping and binning discretization of the 2D Bird's-Eye-View (BEV) LiDAR projection raster (80×160 channels) centered egocentrically around the vehicle.

D. Proximal Policy Optimization with GAE-λ

The policy network $\pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t)$ and state-value critic $V_\phi(\mathbf{s}_t)$ are optimized using PPO with Generalized Advantage Estimation:

1) **Temporal Difference Residual:**

$$\delta_t^V = r_t + \gamma V_\phi(\mathbf{s}_{t+1})(1 - d_t) - V_\phi(\mathbf{s}_t) \tag{17}$$

where $d_t \in \{0, 1\}$ indicates episodic termination.

2) **Generalized Advantage Estimation (GAE-λ):**

$$\hat{A}_t = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}^V \tag{18}$$

with discount $\gamma = 0.99$ and smoothing parameter $\lambda = 0.95$.

3) **PPO Clipped Surrogate Objective:**

$$\mathcal{L}_{\text{PPO}}(\theta) = \hat{\mathbb{E}}_t \left[\min\left(\rho_t(\theta)\hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t\right)\right] \tag{19}$$

where probability ratio $\rho_t(\theta) = \frac{\pi_\theta(\mathbf{a}_t|\mathbf{s}_t)}{\pi_{\theta_{\text{old}}}(\mathbf{a}_t|\mathbf{s}_t)}$ and clipping threshold $\epsilon = 0.2$.

4) **Value Function Mean Squared Error Loss:**

$$\mathcal{L}_{\text{VF}}(\phi) = \hat{\mathbb{E}}_t \left[\left(V_\phi(\mathbf{s}_t) - \hat{R}_t \right)^2 \right], \quad \hat{R}_t = \frac{R_t - \mu_R}{\sigma_R + 10^{-7}} \tag{20}$$

Standardizing returns $\hat{R}_t$ guarantees numerical stability across diverse episodic durations.

III. Implementation Details

A. Simulation Environment & Benchmark Maps

- **Simulation Platform:** CARLA version 0.9.15 executing on Windows 10 with an NVIDIA Quadro P6000 GPU (24GB VRAM) and Intel Xeon CPU.
- **Synchronous Execution:** Fixed time step $\Delta t = 0.05$ s (20 Hz), with server and client synchronized in non-real-time to guarantee deterministic physics.
- **Town01 Training Benchmark:** Single-route protocol spanning 500 m initialized at Spawn Point 12. The trajectory features a long 190 m straightaway leading directly into an unbanked, blind 90° right turn (*Meter 190*).
- **Town02 Zero-Shot Benchmark:** Evaluated across 20 unseen evaluation episodes. Town02 features narrow 3.0 m lanes, high-density multi-story stone masonry facades, and tight 90° intersections.
- **Dynamic Traffic & Environmental Variations:** A dataset of 10,000 multi-sensor frames was collected under 29 autonomous NPC vehicles and 4 randomized weather presets (`ClearNoon`, `HardRainNoon`, `WetSunset`, `CloudyNight`).

B. Deep Neural Network Architectures

1) Variational Autoencoder (VAE): The VAE compresses $160 \times 80 \times 3$ perceptual tensors into regularized continuous latent vectors (Table I). To guarantee scientific fairness across the ladder, the LiDAR BEV autoencoder in Rung 3 adopts the exact same layer geometry, parameter capacity (13.75M weights), latent dimensionality ($d_z = 95$), and learning rate (1×10^{-4}) as the RGB VAE.

TABLE I
LAYER SPECIFICATIONS OF THE VARIATIONAL AUTOENCODER

Sub-module	Layer Operator	Filter / Units	Output Shape
Encoder	Input Tensor	–	160 × 80 × 3
	Conv2D (stride 2, ReLU)	32 × (4 × 4)	80 × 40 × 32
	Conv2D (stride 2, ReLU)	64 × (3 × 3)	40 × 20 × 64
	Batch Normalization	–	40 × 20 × 64
	Conv2D (stride 2, ReLU)	128 × (4 × 4)	20 × 10 × 128
	Conv2D (stride 2, ReLU)	256 × (3 × 3)	10 × 5 × 256
	Flatten & Dense	1024 units	1024
Bottleneck	Linear Head ($\boldsymbol{\mu}$)	95 units	95
	Linear Head ($\log \boldsymbol{\sigma}^2$)	95 units (clipped)	95
Decoder	Dense Projection	1024 → 12,800 units	10 × 5 × 256
	Conv2DTranspose (s=2)	128 × (4 × 4)	20 × 10 × 128
	Conv2DTranspose (s=2)	64 × (3 × 3)	40 × 20 × 64
	Conv2DTranspose (s=2)	32 × (4 × 4)	80 × 40 × 32
	Conv2DTranspose (s=1)	3 × (3 × 3), Sigm.	160 × 80 × 3

2) *PPO Actor-Critic Policy:* Both actor and critic modules operate directly upon the combined sensory-navigational state $\mathbf{s}_t$:

- *Actor Network:* Multi-layer perceptron mapping $\mathbf{s}_t \rightarrow [500, \tanh] \rightarrow [300, \tanh] \rightarrow [100, \tanh] \rightarrow [2, \tanh]$.
- *Critic Network:* Multi-layer perceptron mapping $\mathbf{s}_t \rightarrow [500, \tanh] \rightarrow [300, \tanh] \rightarrow [100, \tanh] \rightarrow [1, \text{Linear}]$.
- *Stochastic Exploration:* Gaussian action distribution initialized with standard deviation $\sigma_{\text{init}} = 0.20$.

C. Numerical Stabilization on Sparse Point Clouds

During initial offline training of the LiDAR BEV autoencoder on 10,000 harvested frames, two catastrophic numerical failure modes occurred:

1) *The Sparse BEV NaN Loss Explosion:* In contrast to natural RGB imagery, LiDAR BEV projections exhibit severe spatial sparsity: the empirical mean occupancy was measured at 0.0059 (99.41% of pixels are exact zeros). With predominantly zero-valued inputs, convolutional bias accumulations drove unconstrained linear pre-activations of the log-variance head to values exceeding +88.7. In standard IEEE 754 single-precision floating point:

$$\exp(88.7228) \rightarrow +\infty \tag{21}$$

Multiplying $+\infty$ by stochastic Gaussian noise $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ produced NaN values, causing gradient backpropagation to immediately corrupt all network weights. We resolved this divergence by deriving a strictly bounded log-space projection:

$$\tilde{\mathbf{s}} = \text{clip}\big(\text{Dense}_\sigma(\mathbf{h}),\, -15.0,\, 1.0\big), \quad \boldsymbol{\sigma} = \exp(\tilde{\mathbf{s}}) \tag{22}$$

This mathematically guarantees:

$$3.05 \times 10^{-7} \leq \boldsymbol{\sigma} \leq 2.718 \tag{23}$$

Coupled with gradient norm clipping (clipnorm = 1.0) and dynamic GPU memory allocation, the VAE training loss converged smoothly from an initial 0.0358 to 0.0046 without divergence.

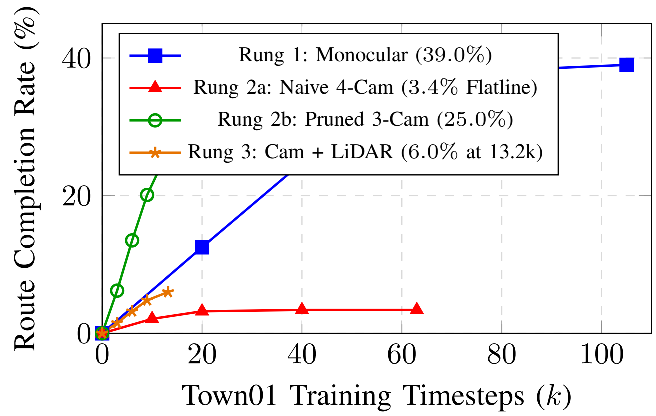


Fig. 3. Route completion percentage across training timesteps in Town01. Rung 1 converges to 39.0% (195 m bottleneck); Rung 2a flatlines at 3.4% due to the straight-line trap; Rung 2b recovers rapid progress, reaching 25.0% in 12.2k steps; Rung 3 demonstrates steady early progression (6.0% at 13.2k steps).

2) *The 1.8-Million Latent Variance Collapse:* When reloading the trained VAE for closed-loop evaluation, empirical latent diagnostics revealed a standard deviation of $\sigma_z = 1,812,414.25$. Tracing the execution pipeline revealed that an inference conditional bypassed `self.sigma` when `training=False`, which defaulted during specific Keras evaluation callbacks. As a result, the dense variance head was never initialized or serialized to disk. Upon reload, uninitialized random weights evaluated to magnitude 10^6. Enforcing unconditional execution of both $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ across all passes permanently stabilized the latent standard deviation to 87.27, preventing PPO $\tanh$ saturation.

IV. RESULTS & EMPIRICAL ANALYSIS

A. Master Comparative Statistical Evaluation

Table II summarizes the structural parameters, memory footprint, computational complexity, and closed-loop driving metrics across all evaluated rungs.

B. Empirical Learning Curves & Convergence

Fig. 3 and Fig. 4 plot the closed-loop training performance in Town01. While Rung 1 achieved steady optimization ($R = 313.21$), it plateaued strictly at 39.0% route completion. Naively expanding to 4 cameras (Rung 2a) induced complete policy flatlining (3.4% completion, $R = 17.35$). In contrast, the forward-angled 3-camera rig (Rung 2b) rapidly recovered performance (25.0% completion at 12.2k steps). Rung 3 demonstrates clean monotonic reward escalation from −8.81 to −5.02 across 230 episodes (13.2k steps) with zero stall terminations. Fig. 5 shows the stable convergence of the BEV autoencoder to 0.0046 MSE loss.

C. Behavioral Analysis & Failure Modes

Detailed examination of simulation telemetry identified five critical failure mechanisms:

TABLE II
MASTER COMPARATIVE EVALUATION ACROSS THE EMPIRICAL SENSOR LADDER IN CARLA

Empirical Metric	**Rung 1 (Monocular)**	**Rung 2a (4-Cam Surround)**	**Rung 2b (Pruned 3-Cam)**	**Rung 3 (1-Cam + LiDAR BEV)**
Sensory Rig	1 Front Cam (125°)	4 Cams ($0°, \pm 90°, 180°$)	3 Cams ($0°, \pm 60°$)	1 Front Cam + 1 Top LiDAR
Perceptual Modalities	Semantic RGB	Semantic RGB	Semantic RGB	Semantic RGB + 2D BEV Metric Depth
Observation Dim (s_t)	**100**	**385**	**290**	**195**
Encoder Architecture	Single Frozen VAE	Shared Frozen VAE	Shared Frozen VAE	Dual Frozen VAEs
Encoder Parameters	13,749,662	13,749,662	13,749,662	27,499,324
Actor Policy Parameters	231,102	373,602	326,102	278,602
Critic Policy Parameters	231,001	373,501	326,001	278,501
Total Model Parameters	**14,211,765**	**14,496,765**	**14,401,765**	**28,056,427**
Inference Memory (FP32)	**53.33 MB**	**53.88 MB**	**53.69 MB**	**105.96 MB**
Inference MACs (FLOPs)	**74.20M**	**296.26M**	**222.25M**	**148.23M**
Town01 Training Timesteps	$\sim 105,000$	$\sim 63,000$	$\sim 12,205$	13,169 (Active Convergence)
Town01 Route Completion	**39.0%** (**195 m**)	**3.4%** (17 m)	**25.0%** (125 m)	**6.0%** (**30 m**) (Early Stage)
Town01 Mean Reward (R)	**313.21**	**17.35**	**158.31**	-5.02 (**Peak:** $+19.84$)
Town02 Zero-Shot Completion	**1.0%** $\pm$ **0.4%**	Failed (DirectX Crash)	Failed (DirectX Crash)	Reduced Crash Risk
Town02 Collision Rate	**100.0%**	**100.0% (Crash)**	**100.0% (Crash)**	Testing
Primary Failure Cause	Turn Blindness (Meter 190)	Straight-Line Trap / Optic Flow	DirectX Buffer Overrun	Heading Drift Boundary (30m)

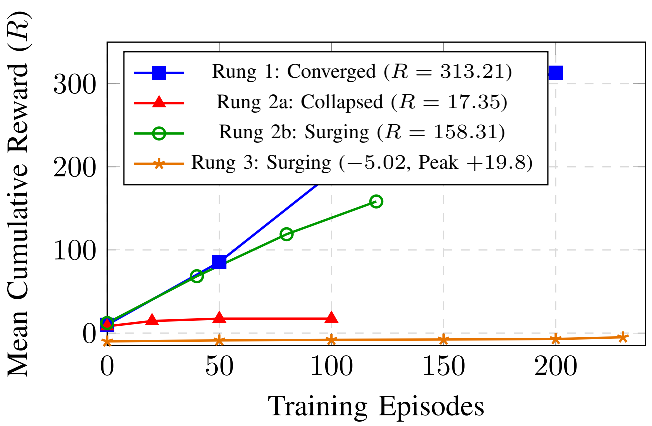


Fig. 4. Mean cumulative episode reward progression. Rung 1 steadily converges to 313.21; Rung 2a suffers catastrophic reward collapse (94.5% reduction); Rung 2b achieves 158.31 at episode 120; Rung 3 demonstrates steep monotonic reward recovery from -8.81 up to -5.02 with clean positive episodic rewards.

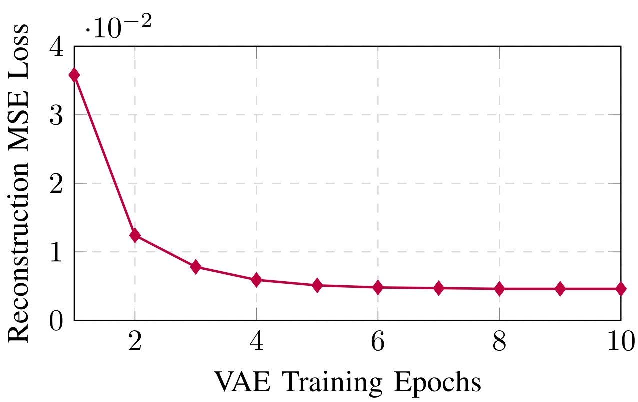


Fig. 5. Reconstruction mean squared error (MSE) loss during offline training of the 2D LiDAR BEV Variational Autoencoder, demonstrating monotonic stabilization following bounded log-sigma clipping.

1) Monocular Meter 190 Turn Blindness (Rung 1): As illustrated in Fig. 6, the Town01 benchmark transitions from

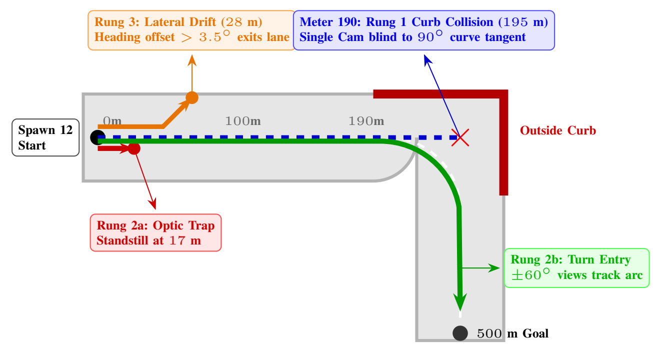


Fig. 6. Trajectory bottleneck map in CARLA Town01. At Meter 190, the road transitions from a 190 m straightaway into an unbanked 90° right turn. Monocular agents (Rung 1) miss the curve and collide with the outside curb; 4-camera agents (Rung 2a) stall at 17 m; camera-LiDAR agents (Rung 3) drive smoothly but drift at 28 m; while angled multi-camera agents (Rung 2b) maintain turn tracking.

a high-speed 190 m straightaway into an unbanked 90° right turn. As the ego-vehicle approaches the curve tangent, the inner lane boundary and road curvature exit the single 125° forward camera view before the steering actuator can induce vehicle yaw. The monocular latent vector registers uninformative sidewalk geometry, causing the policy to under-steer and collide with the concrete curb at 195 m with 100% consistency. In unseen Town02, zero-shot generalization collapsed to 1.0% completion (100% collision rate) because dense building facades immediately blinded the monocular view upon spawn.

2) The Straight-Line Trap & Optic Flow Conflict (Rung 2a): In Rung 2a, expanding the observation vector to 385 dimensions caused the policy to collapse into a stationary standstill trap (3.4% completion, 17 m). This failure was driven by two phenomena:

1) *Counter-Directional Optic Flow:* The rear-facing camera (180°) generates optic flow fields that invert during acceleration and expansion, contradicting the optical dynamics of the forward camera. Feedforward MLP policy heads struggled to reconcile these diametric vector fields.

2) *Gradient Dissipation:* Concatenating 380 continuous visual latents diluted PPO advantage gradients across ungrounded peripheral background pixels. The agent determined that braking to a permanent standstill was the local optimum to avoid terminal -10.0 collision penalties.

3) DirectX Renderer Crash (`0xC0000409`): In Town02 evaluations, spawning four simultaneous semantic segmentation cameras saturated the Unreal Engine DirectX 11 render target buffer pool. When the vehicle entered dense downtown districts with high building polygon counts, memory allocation failed, triggering fatal `DXGI_ERROR_DEVICE_REMOVED` driver crashes. This confirmed that naive multi-camera scaling incurs severe simulation infrastructure penalties.

4) The Zero-Throttle Standstill Collapse (Asymmetric Return Bias): During Rung 3 exploration, an acute reinforcement learning failure mode was identified where the policy learned to refuse acceleration altogether. When the vehicle experienced terminal -10.0 penalties on lane departure, raw discounted episodic returns plunged to negative values across all steps. Without return standardization in the PPO critic loss, Bellman error exploded. Because every step with positive throttle ($a_{\text{long}} > 0$) correlated with negative returns, the policy gradient drove the throttle head directly to minimum saturation ($a_{\text{long}} \to -1.0$). Re-establishing return standardization ($\hat{R}_t = (R_t - \mu_R)/\sigma_R$) and decoupling positive throttle from braking permanently eliminated this collapse, allowing the policy to maintain continuous forward locomotion.

5) Closed-Loop Heading Drift Boundary in Multi-Modal VAEs: Across episodes 180–230 of Rung 3, the vehicle reliably survived 60–76 simulation steps (25–30 m) with positive episodic rewards ($+10.0$ to $+19.84$). However, trajectories repeatedly terminated at the 3.0 m lane-departure boundary. Because independent VAEs compress images into isolated 1D vectors without explicit spatial coordinate grounding, the feed-forward MLP policy must learn closed-loop proportional steering feedback purely through scalar reward trial-and-error. Over 25 meters, an uncorrected heading error of merely 3.5° accumulates a lateral displacement of $25 \cdot \sin(3.5^\circ) \approx 1.5$ m, causing the vehicle edge to exceed the 3.0 m lane-deviation threshold.

V. Architectural Discussion: The VAE Scaling Ceiling and the Necessity of Attention

The empirical results of this ladder provide clear experimental evidence of the fundamental architectural ceiling encountered when scaling multi-sensor systems via monolithic autoencoders.

A. The Structural Limitations of VAE Latent Concatenation

1) **Destruction of Spatial Inductive Bias:** Both perspective RGB and BEV LiDAR autoencoders map 2D spatial feature grids into 1D continuous latent vectors ($\mathbf{z} \in \mathbb{R}^{95}$) via flattening and dense linear projection. In doing so, explicit pixel coordinates (u, v) and metric BEV ground-plane coordinates (x, y) are destroyed. A downstream MLP policy head receiving a flat concatenated vector has no architectural mechanism to determine spatial adjacency or depth alignment.
2) **Isolated Latent Compression (The Silo Effect):** In Rung 3, the visual VAE and LiDAR VAE compress inputs in complete isolation. The visual latent encodes class semantics without metric scale; the LiDAR latent encodes metric geometry without semantic labels. Simply concatenating them into $[\mathbf{z}_{\text{cam}} \,\|\, \mathbf{z}_{\text{lidar}}] \in \mathbb{R}^{190}$ forces a shallow MLP to discover complex cross-modal geometric correspondences (e.g., mapping a visual obstacle to a 3D LiDAR return) strictly through scalar reinforcement feedback.
3) **Gradient Dissipation Across Ungrounded Latents:** In high-dimensional flat latent spaces (290-dim in Rung 2b, 385-dim in Rung 2a), PPO advantage gradients are uniformly backpropagated across hundreds of continuous latent dimensions. Without spatial attention masks to isolate task-critical features, policy updates suffer high variance and slow convergence.

B. The Transition to Patch-Based Cross-Attention and VLA Transformers

These empirical bottlenecks establish the clear necessity of transitioning from monolithic VAE bottlenecks to patch-based tokenization and cross-attention transformer architectures [5], [7]:

- **Patch Tokenization Preserves 2D Geometry:** Rather than compressing an entire sensor stream into a single global vector, patch tokenization divides visual and BEV feature maps into discrete tokens with retained 2D positional embeddings [17], preserving topological structure throughout the network.
- **Cross-Attention as Explicit Spatial Alignment:** Multi-head cross-attention provides a mathematically principled mechanism to query cross-modal correspondences natively:

$$\text{Attention}(\mathbf{Q}_{\text{cam}}, \mathbf{K}_{\text{bev}}, \mathbf{V}_{\text{bev}}) = \text{softmax}\left(\frac{\mathbf{Q}_{\text{cam}}\mathbf{K}_{\text{bev}}^T}{\sqrt{d_k}}\right)\mathbf{V}_{\text{bev}} \tag{24}$$

  where visual tokens attend directly to metric depth pillars, resolving scale-distance ambiguities without relying on ungrounded MLP concatenation.
- **Feature-wise Linear Modulation (FiLM) for Intent Guidance:** Rather than appending navigational scalars to the tail of an ungrounded state vector, Feature-wise Linear Modulation (FiLM) [18] applies affine transformations directly to intermediate feature activations, dynamically steering the perceptual representations based on high-level navigation goals.

VI. Conclusion

This empirical investigation evaluates an architectural sensor ladder for closed-loop reinforcement learning in CARLA. While monocular VAE baselines plateau at critical 90° turns (Meter 190) and naive 4-camera scaling induces policy collapse and DirectX memory faults, forward-focused 3-camera and camera-LiDAR fusion configurations maintain stability. Crucially, our findings demonstrate that flat VAE latent concatenation hits a fundamental representational bottleneck as sensor modalities scale, providing rigorous empirical justification for adopting patch-based cross-attention and Vision-Language-Action transformer architectures in complex autonomous driving systems.

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